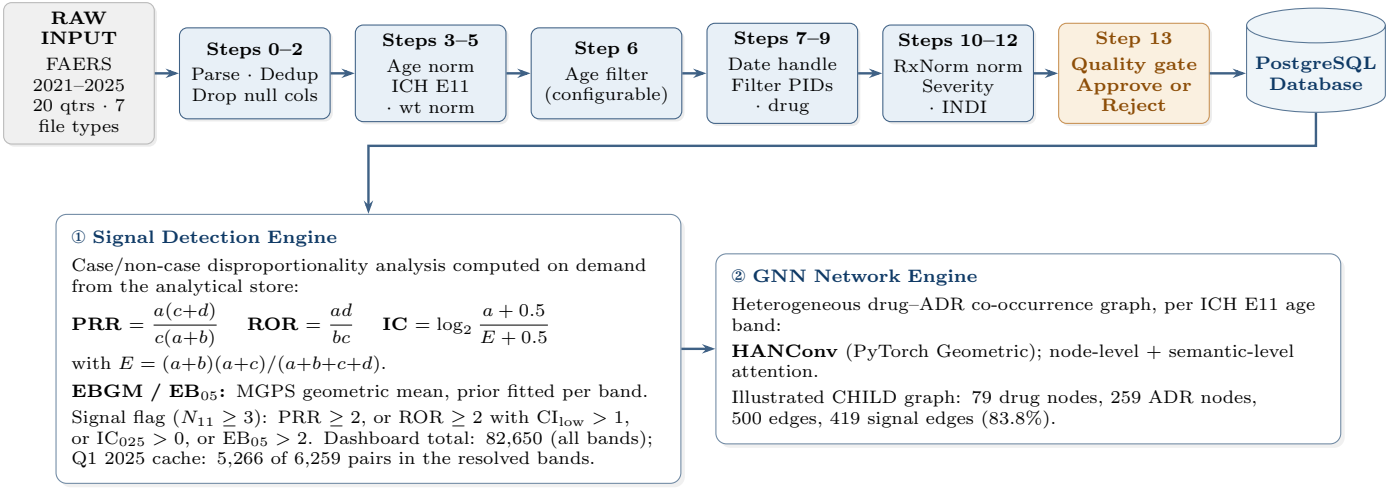

RAW INPUT
FAERS 2021–2025 20 qtrs · 7 file types
Steps 0–2
Parse · Dedup Drop null cols
Steps 3–5
Age norm ICH E11 · wt norm
Step 6
Age filter (configurable)
Steps 7–9
Date handle Filter PIDs · drug
Steps 10–12
RxNorm norm Severity · INDI
Step 13
Quality gate Approve or Reject
PostgreSQL Database
① Signal Detection Engine
Case/non-case disproportionality analysis computed on demand from the analytical store:
PRR = a(c+d)/c(a+b) ROR = ad/bc IC = log2 (a + 0.5)/(E + 0.5)
with E = (a+b)(a+c)/(a+b+c+d).
EBGM / EB05: MGPS geometric mean, prior fitted per band.
Signal flag (N11 ≥ 3): PRR ≥ 2, or ROR ≥ 2 with CIlow > 1, or IC025 > 0, or EB05 > 2. Dashboard total: 82,650 (all bands); Q1 2025 cache: 5,266 of 6,259 pairs in the resolved bands.
② GNN Network Engine
Heterogeneous drug–ADR co-occurrence graph, per ICH E11 age band:
HANConv (PyTorch Geometric); node-level + semantic-level attention.
Illustrated CHILD graph: 79 drug nodes, 259 ADR nodes, 500 edges, 419 signal edges (83.8%).